

DAEN/ISEN 427 Homework Assignment

Salary is negotiable, take or leave it

Due: Thu Nov 20 9:35 AM

Name: _____

Points: _____

Instructions: Read ... answer the **blue** questions (in bullet points or full sentences). Understanding strategic behavior can help engineers negotiate more effectively. Explore **game theory concepts** such as Nash equilibrium, payoff matrices, and strategic decision-making to job negotiations.

Two players are engaged in a one-shot negotiation game:

- **Player A (engineer):** negotiating job terms.
- **Player B (employer):** deciding salary offer.

Each decides simultaneously between two strategies:

Engineer Choices: Accept or Reject the offer

Employer Choices: Offer High Salary or Low Salary

Payoff Matrix:

	Employer: High Offer	Employer: Low Offer
Engineer: Accept	(4, 4) Stable hire	(1, 5) Unfair deal
Engineer: Reject	(2, 0) Employer loses	(0, 1) Both lose

Payoffs are relative satisfaction values (Engineer, Employer).

Questions:

1. Identify any **dominant strategies** for each player. [1pt]
2. Determine the **Nash equilibrium** of the game. [1pt]
3. How **information asymmetry** (e.g., uncertain market salaries) might alter the equilibrium? [1pt]
4. Consider a **repeated game**: How might negotiation strategies evolve over multiple rounds? [1pt]

Additional Questions (for your reflexion, no need to answer):

- 5 Adjust payoffs to represent **risk-averse** or **risk-seeking** behavior.
- 6 Model different labor market conditions (e.g., high demand vs. job scarcity).
- 7 Present outcomes using a **decision tree** or utility graph.

Bayesian inference can help you **update your expectations** as new information becomes available. In a negotiation or job interview, you begin with a belief (your *prior*) about the employer's intentions, and you revise it when you get new evidence (their responses, tone, or offer details).

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

- $P(H)$ = Prior belief (what you initially think, e.g., chance the employer will accept your counteroffer)
- $P(E|H)$ = Likelihood (how likely their response is, assuming your belief is true)
- $P(E)$ = Overall probability of the response
- $P(H|E)$ = Posterior belief (your updated belief after observing their behavior)

Job Interview Scenario: Suppose an engineer estimates there is a **50% chance** that a company will accept a higher salary request (**prior**). During the interview, the employer says: *"We're open to negotiation for strong candidates."* From experience, engineers know that when a company is truly flexible, such a statement occurs 80% of the time. When a company is not flexible, it still occurs 30% of the time (they often say it politely anyway).

$$P(\text{Flexible}|\text{Statement}) = \frac{0.80 \times 0.50}{0.80 \times 0.50 + 0.30 \times 0.50} = \frac{0.40}{0.55} \approx 0.73$$

So after hearing the comment, the engineer updates their belief: **the chance the employer will negotiate fairly rises from 50% to about 73%.**

Bayesian inference in job negotiation is like adjusting your confidence level:

- You start with an assumption (prior),
- You get a signal during the conversation (evidence),
- You update your expectation (posterior) before making your next move.

$$\text{New Belief} = \text{Old Belief} \times \text{How well the employer's behavior fits your expectations}$$

- 8 In your words, how can an engineering candidate use Bayesian reasoning to make better strategic decisions during interviews? Briefly discuss how observing the employer's language, tone, or behavior can update the candidate's beliefs and change their optimal strategy in a negotiation game. [1pt]